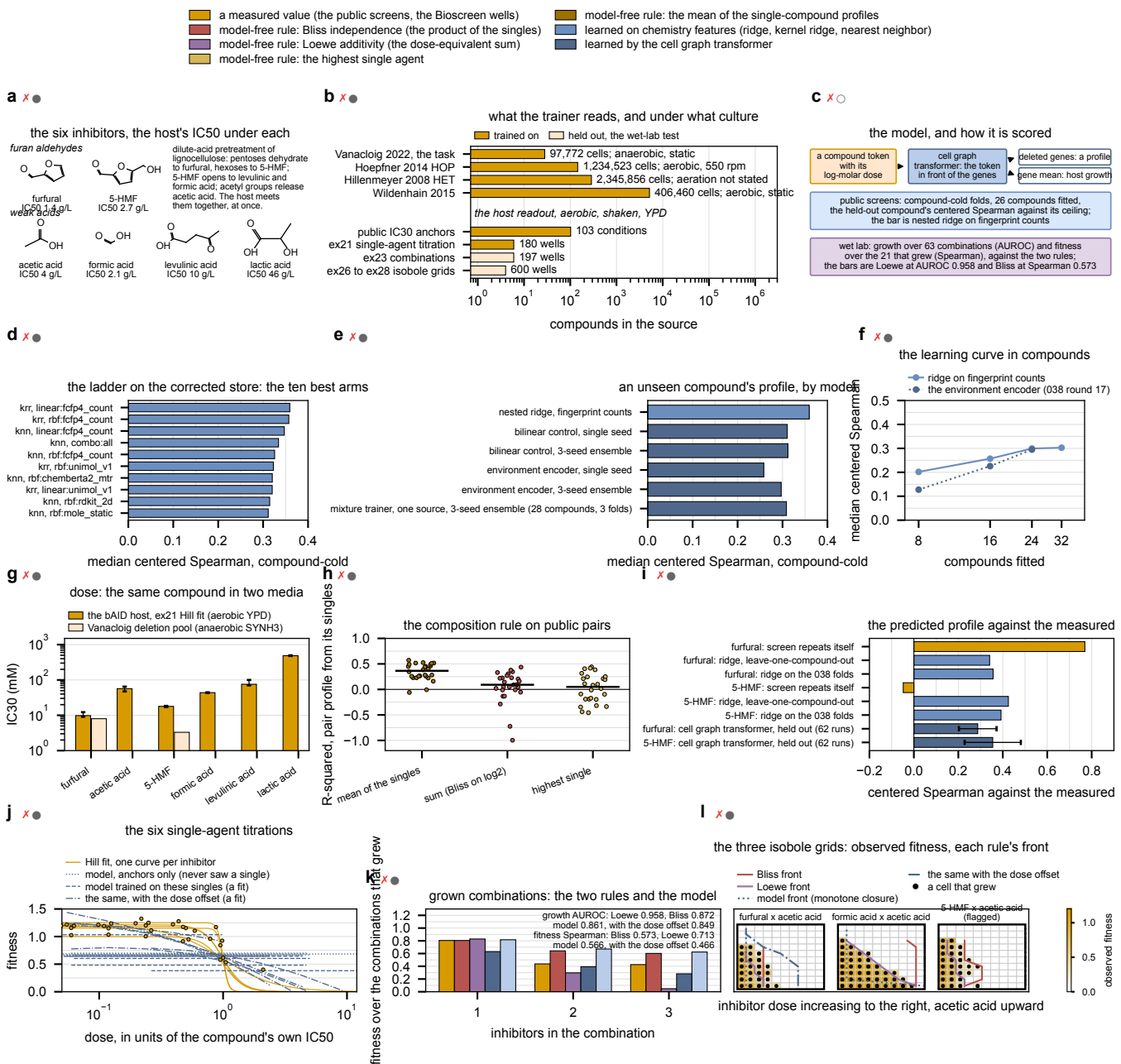

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the six inhibitors, the host's IC50 under each
furan aldehydes
furfural IC50 1.4 g/L
5-HMF IC50 2.7 g/L
dilute-acid pretreatment of lignocellulose: pentoses dehydrate to furfural, hexoses to 5-HMF; 5-HMF opens to levulinic and formic acid; acetyl groups release acetic acid. The host meets them together, at once.
weak acids
acetic acid IC50 4 g/L
formic acid IC50 2.1 g/L
levulinic acid IC50 10 g/L
lactic acid IC50 46 g/L
b
what the trainer reads, and under what culture
trained on
held out, the wet-lab test
Vanacloig 2022, the task
Hoepfner 2014 HOP
Hillenmeyer 2008 HET
Wildenhain 2015
97,772 cells; anaerobic, static
1,234,523 cells; aerobic, 550 rpm
2,345,856 cells; aeration not stated
406,460 cells; aerobic, static
the host readout, aerobic, shaken, YPD
public IC30 anchors
ex21 single-agent titration
ex23 combinations
ex26 to ex28 isobole grids
103 conditions
180 wells
197 wells
600 wells
compounds in the source
c
the model, and how it is scored
a compound token with its log-molar dose
cell graph transformer: the token in front of the genes
deleted genes: a profile
gene mean: host growth
public screens: compound-cold folds, 26 compounds fitted, the held-out compound's centered Spearman against its ceiling; the bar is nested ridge on fingerprint counts
wet lab: growth over 63 combinations (AUROC) and fitness over the 21 that grew (Spearman), against the two rules; the bars are Loewe at AUROC 0.958 and Bliss at Spearman 0.573
d
the ladder on the corrected store: the ten best arms
krr, linear:fcfp4_count
krr, rbf:fcfp4_count
knn, linear:fcfp4_count
knn, combo:all
knn, rbf:fcfp4_count
krr, rbf:unimol_v1
knn, rbf:chemberta2_mtr
krr, linear:unimol_v1
knn, rbf:rdkit_2d
knn, rbf:mole_static
median centered Spearman, compound-cold
e
an unseen compound's profile, by model
nested ridge, fingerprint counts
bilinear control, single seed
bilinear control, 3-seed ensemble
environment encoder, single seed
environment encoder, 3-seed ensemble
mixture trainer, one source, 3-seed ensemble (28 compounds, 3 folds)
median centered Spearman, compound-cold
f
the learning curve in compounds
ridge on fingerprint counts
the environment encoder (038 round 17)
median centered Spearman
compounds fitted
g
dose: the same compound in two media
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
h
the composition rule on public pairs
R-squared, pair profile from its singles
mean of the singles
sum (Bliss on log2)
highest single
i
the predicted profile against the measured
furfural: screen repeats itself
furfural: ridge, leave-one-compound-out
furfural: ridge on the 038 folds
5-HMF: screen repeats itself
5-HMF: ridge, leave-one-compound-out
5-HMF: ridge on the 038 folds
furfural: cell graph transformer, held out (62 runs)
5-HMF: cell graph transformer, held out (62 runs)
centered Spearman against the measured
j
the six single-agent titrations
Hill fit, one curve per inhibitor
model, anchors only (never saw a single)
model trained on these singles (a fit)
the same, with the dose offset (a fit)
fitness
dose, in units of the compound's own IC50
k
grown combinations: the two rules and the model
growth AUROC: Loewe 0.958, Bliss 0.872
model 0.861, with the dose offset 0.849
fitness Spearman: Bliss 0.573, Loewe 0.713
model 0.566, with the dose offset 0.466
fitness over the combinations that grew
inhibitors in the combination
l
the three isobole grids: observed fitness, each rule's front
Bliss front
Loewe front
model front (monotone closure)
the same with the dose offset
a cell that grew
furfural x acetic acid
formic acid x acetic acid
5-HMF x acetic acid (flagged)
observed fitness
inhibitor dose increasing to the right, acetic acid upward